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MEC 413 — Past Questions & Materials

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Paper 1 of 1 — Operation Management MEC 413 lecture notes

Level HND2 • First Semester • Handout

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PETROLEUM TRAINING INSTITUTE, EFFURUN

LECTURE NOTE ON

OPERATIONS MANAGEMENT MEC 413

OUTLINE:

1. INTRODUCTION TO OPTS MGT
2. COMPETITIVE PROFILE
3. CAPACITY PLANNING
4. MATERIAL MANAGEMENT
5. SCHEDULING
6. LAYOUT PLANNING
7. JUST-IN-TIME (JIT) PRINCIPLES
8. LEAN AND AGILE MANUFACTURING

Lecture Note on MEC 413 By Engr. A. C. ANEKE

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1.0 OPERATIONS MANAGEMENT

Production is the creation of goods and services. Operations Management is the set of activities that creates value in the form of goods and services by transforming inputs into outputs. Activities creating goods and services take place in all organizations. In manufacturing firms, the production activities that create goods are usually quite obvious. In them, we can see the creation of a tangible product such as Indomie nodules or Innoson plastic.

In organizations that do not create physical products the production function may be less obvious. It may be hidden from the public and even from the customer. Examples are the transformations that take place at a bank, hospital, airline office or University.

Often when services are performed, no tangible goods are produced. Instead, the product may take such forms as the transfer of funds from a savings account to a checking account, the transplant of a liver, the filling of an empty seat on an airline, or the education of a student. Regardless of whether the end product is a good or service, the production activities that go on in the organization are often referred to as operation or operations management.

1.1 Organizing to Produce Goods and Services

To create goods and services, all organizations perform three functions. These functions are the necessary ingredients not only for production but also for an organization's survival. They are

1. Marketing: which generates the demand, or at least, takes the order for a product or service (nothing happens until there is a sale).
2. Production/Operations, which creates the product.
3. Finance/Accounting, which tracks how well the organization is doing, pays the bills, and collects the money.

1.2 Why Study Operations Management

1. Operations Management is one of the three major functions of any organization, and it is integrally related to all the other business functions. All organizations market, finance, and produce, and it is important to know how the Operations Management activity functions. Therefore we study how people organise themselves for productive enterprise.
2. We study OM because we want to know how goods and services are produced. The production function is the segment of our society that creates the products we use.
3. We study OM to understand what operations managers do. By understanding what these managers do, you can develop the skills

necessary to become such a manager. This will help you explore the numerous and lucrative career opportunities in Operations Management.

4. We study OM because it is such a costly part of an organisation. A large percentage of the revenue of most firms is spent in the OM function. Indeed, OM provides a major opportunity for an organization to improve its profitability and enhance its service to society.

1.3 What Operations Managers Do

All good managers perform the basic functions of the management process. The management process consists of planning, organizing, staffing, leading, and controlling. Operations managers apply this management process to the decisions they make in the OM function.

2.0 COMPETITIVE PROFILE

Competitive profile in management is an assessment of the strengths and weaknesses of current and potential competitors. This analysis provides both an offensive and defensive strategic context to identify opportunities and threats. Profiling coalesces (blends) all of the relevant sources of competitor analysis into one framework in the support of efficient and effective strategy formulation, implementation, monitoring and adjustment.

Competitive profile is an essential component of corporate strategy. It is argued that most firms do not conduct this type of analysis systematically enough. Instead, many enterprises operate on what is called "informal impressions, conjectures, and intuition gained through the tidbits of information about competitors every manager continually receives." As a result, traditional environmental scanning places many firms at risk of dangerous competitive blind spots due to a lack of robust competitor analysis.

2.1 Competitive Advantage

Competitive advantage is a business concept describing attributes that allow an organization to outperform its competitors. These attributes may include access to natural resources, such as high grade ores or inexpensive power, highly skilled personnel, geographic location, high entry barriers, etc. New technologies, such as robotics and information technology, can also provide competitive advantage, whether as a part of the product itself, as an advantage to the making of the product, or as a competitive aid in the business process (for example, better identification and understanding of customers).

In other words, competitive advantage may be said to be the creation of a system that has a unique advantage over its competitors. The idea is to create customer value in an efficient and sustainable way. Managers achieve competitive advantage via differentiation, low cost and response.

2.1-1 Competing on Differentiation

Differentiation is concerned with providing uniqueness. A firm's opportunities for creating uniqueness are not located within a particular function or activity but can arise in virtually everything that the firm does. Moreover, because most products include some service, and most services include some product, the opportunities for creating this uniqueness are limited only by

imagination. Indeed, differentiation should be thought of as going beyond both physical characteristics and service attributes to encompass everything about the product or service that influences the value that the customers derive from it. Therefore, effective operation managers assist in defining everything about a product or service that will influence the potential value to the customer. This may be convenience of a broad product line, product features, or a service related to the product. Such services can manifest themselves through convenience (location of distribution centres or stores), training, product delivery and installation, or repair and maintenance services.

2.1-2 Competing on Cost

One driver of a low-cost strategy is a facility that is effectively utilized. Identifying the optimum size (and investment) allows firms to spread overhead costs, providing a cost advantage.

Low – cost leadership entails achieving maximum value as defined by your customer. It requires a relentless effort to drive down costs while meeting customer expectations of value. A low – cost strategy does not imply low value or low quality.

2.1-3 Competing on Response

The third strategy option is response. Response is often thought of as flexible response, but it also refers to reliable and quick response. Indeed, we define response as including the entire range of values related to timely product development and delivery, as well as reliable scheduling and flexible performance.

Flexible response may be thought of as the ability to match changes in a marketplace where design innovations and volumes fluctuate substantially.

2.2 Competitive Profile Matrix (CPM)

This is the key question in analysing the competition.

The *Competitive Profile Matrix (CPM)* is a tool that compares an organization and its rivals and reveals their relative strengths and weaknesses.

Understanding the tool

In order to better understand the external environment and the competition in a particular industry, firms often use CPM. The matrix identifies a firm's key competitors and compares them using industry's critical success factors. The

analysis also reveals company's relative strengths and weaknesses against its competitors, so a company would know, which areas it should improve and, which areas to protect. An example of a matrix is demonstrated below.

CPM Table

Critical Success Factor	Company A Company B Company C					
	Weight	Rating	Score	Rating	Score	Rating
Brand reputation	0.13	2	0.26	3	0.39	1
Level of product integration	0.08	4	0.32	3	0.24	1
Range of products	0.05	3	0.15	1	0.05	2
Successful new introductions	0.04	3	0.12	3	0.12	3
Market Share	0.14	2	0.28	4	0.56	4
Sales per employee	0.08	1	0.08	2	0.16	3
Low cost structure	0.05	1	0.05	3	0.15	4
Variety of distribution channels	0.07	4	0.28	2	0.14	2
Customer retention	0.02	2	0.04	4	0.08	1
Superior IT capabilities	0.11	3	0.33	4	0.44	4
Strong online presence	0.15	3	0.45	3	0.45	4
Successful promotions	0.08	1	0.08	2	0.16	1
Total	1.00	-	2.44	-	2.94	-

Critical Success Factors

Critical success factors (CSF) are the key areas, which must be performed at the highest possible level of excellence if organizations want to succeed in the particular industry. They vary between different industries or even strategic groups and include both internal and external factors. In the example above, we have included 12 CSF, which is usually not enough. The more critical success factors are included the more robust and accurate the analysis is. The following list provides some of the general CSF, but the list is not definite and you should include industry specific factors in your matrix:

Market Share	Union relations	Power over suppliers
Product Quality	Skilled workforce	Access to key suppliers
Clear strategic direction	Location of facilities	Efficient supply chain

Customer service	Production capacity	Supply chain integration
Customer loyalty	Added product features	On time delivery
Brand reputation	Price competitiveness	Strong online presence
Customer satisfaction	Low cost structure	Effective social media management
Financial position	Variety of products	Experience and skills in e-commerce
Cash reserves	Complementary products	Management qualification and experience
Profit margin	Level of product integration	Innovation in products and services
Inventory turnover	Successful product promotions	Innovative culture
Employee retention	Superior marketing capabilities	Efficient production
Income per employee	Superior advertising capabilities	Lean production system
Innovations per employee	Superior IT capabilities	Strong supplier network
Cost per employee	Size of advertising budget	Strong distribution network
R&D spending	Effectiveness of sales distribution	Product design
Strong patent portfolio	Employee satisfaction	Level of vertical integration
New patents per year	Effective planning and budgeting	Effective corporate social responsibility programs
Revenue per new product	Variety of distribution channels	Sales per outlet
Successful new introductions	Power over distributors	Parent company support

Weight

Each critical success factor should be assigned a weight ranging from 0.0 (low importance) to 1.0 (high importance). The number indicates how important the factor is in succeeding in the industry. If there were no weights

assigned, all factors would be equally important, which is an impossible scenario in the real world. The sum of all the weights must equal 1.0. Separate factors should not be given too much emphasis (assigning a weight of 0.3 or more) because the success in an industry is rarely determined by one or few factors. In our first example, the most significant factors are 'strong online presence' (0.15), 'market share' (0.14), 'brand reputation' (0.13).

Rating

The ratings in CPM refer to how well companies are doing in each area. They range from 4 to 1, where 4 means a major strength, 3 – minor strength, 2 – minor weakness and 1 – major weakness. Ratings, as well as weights, are assigned subjectively to each company, but the process can be done easier through benchmarking. Benchmarking reveals how well companies are doing compared to each other or industry's average. Just remember that firms can be assigned equal ratings for the same factor. For example, if Company A, Company B and Company C, have the market share of 25%, 27% & 28% accordingly, they would all receive the rating of 4 rather than receiving ratings 2, 3 & 4.

Score & Total Score

The score is the result of weight multiplied by rating. Each company receives a score on each factor. Total score is simply the sum of all individual score for the company. The firm that receives the highest total score is relatively stronger than its competitors. In our example, the strongest performer in the market should be Company B (2.94 points).

Benefits of the CPM:

- I. The same factors are used to compare the firms. This makes the comparison more accurate.
- II. The analysis displays the information on a matrix, which makes it easy to compare the companies visually.
- III. The results of the matrix facilitate decision-making. Companies can easily decide which areas they should strengthen, protect or what strategies they should pursue.

Using the tool

Step 1. Identify the critical success factors

To make it easier, use our list of CSF and include as many factors as possible. In addition, the following questions should be helpful in identifying industry's CSF:

- Why consumers prefer Company A over Company B or vice versa?
- What resources, capabilities and competences firms possess?
- What sustainable competitive advantages companies have in the industry?
- Why some companies succeed and others fail in the industry?

Step 2. Assign the weights and ratings

The best way to identify what weights should be assigned to each factor is to compare the best and worst performing companies in the industry. Well performing companies will usually undertake activities that are significant for success in the industry. They will put most of their resources and energy into those activities as compared to low performing organizations. Weights can also be determined in discussion with other top-level managers.

Ratings should be assigned using benchmarking or during team discussions.

Step 3. Compare the scores and take action

You should compare the scores on each factor to identify where company's relative strengths and weaknesses are. In our first example, Company A had relative strength in 'level of product integration', 'product range' and 'variety of distribution channels'. Therefore, Company A should protect these areas while trying to improve its weaknesses in 'sales per employee' and 'market share'.

The company should also improve its strategy to become more successful in the industry.

Example

This is competitive profile matrix example of smartphones operating systems. The main competitors: Google's Android OS, Apple's iOS and Microsoft's Windows Phone operating systems will be compared with each other to find out their relative strengths and weaknesses.

Critical Success Factor	Android OS				iOS				Windows Phone			
	Weight	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Score
Market share	0.13	4	0.52	2	0.26	2	0.26	2	0.26	2	0.26	0.26
Number of apps in store	0.10	4	0.40	4	0.40	2	0.20	2	0.20	2	0.20	0.20
Frequency of updates	0.06	3	0.18	4	0.24	2	0.12	2	0.12	2	0.12	0.12
Design	0.07	3	0.21	3	0.21	3	0.21	3	0.21	3	0.21	0.21
Product brand reputation	0.05	3	0.15	3	0.15	2	0.10	2	0.10	2	0.10	0.10
Distribution channels	0.11	4	0.44	2	0.22	3	0.33	3	0.33	3	0.33	0.33
Usability	0.11	3	0.33	3	0.33	3	0.33	3	0.33	3	0.33	0.33
Customization features	0.04	4	0.16	2	0.08	2	0.08	2	0.08	2	0.08	0.08
Marketing capabilities	0.04	2	0.08	4	0.16	2	0.08	2	0.08	2	0.08	0.08
Company brand reputation	0.10	4	0.40	4	0.40	3	0.30	3	0.30	3	0.30	0.30
Openness	0.02	4	0.08	2	0.04	2	0.04	2	0.04	2	0.04	0.04
Cloud integration	0.12	4	0.48	2	0.24	2	0.24	2	0.24	2	0.24	0.24
Rate of OS crashes	0.08	1	0.08	4	0.32	3	0.24	3	0.24	3	0.24	0.24
Total	1.00	-	3.51	-	3.05	-	2.53	-	2.53	-	2.53	2.53

The CPM analysis reveals that Android is the strongest player in the industry with relative strengths in market share, distribution channels, customization features, openness and cloud integration. On the other hand, iOS prevails in frequency updates, marketing capabilities and the rate of OS crashes. Windows Phone is the weakest of them all and doesn't have any relative strengths against its rivals. The companies should create their strategies according to their strengths and weakness and improve their ratings in the most significant industry's areas.

Critical Success Factors	Weight	Dell		Gateway		Apple	
		Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Product Quality	0.12	3	0.36	3	0.36	3	0.36
Management Experience	0.04	2	0.08	2	0.08	4	0.16
Market Share	0.11	4	0.44	3	0.33	2	0.22
Inventory System	0.09	4	0.36	2	0.18	2	0.18
Price Competitive	0.07	3	0.21	4	0.28	1	0.07
Innovation	0.06	2	0.12	3	0.18	3	0.18
Advertising	0.05	3	0.15	2	0.10	2	0.10
Brand Name	0.08	3	0.24	3	0.24	2	0.16
Market Expansion	0.10	4	0.40	3	0.30	2	0.20
Customer Service	0.05	3	0.15	2	0.10	2	0.10
Financial Position	0.14	4	0.56	3	0.42	3	0.42
Sales Distribution	0.09	3	0.27	2	0.18	2	0.18
Total	1.00		3.34		2.73		2.33

Key Success Factors	Weight	Coca-Cola		Pepsi		Cadbury Schweppes	
		Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Financial Position	0.08	4	0.32	3	0.24	3	0.24
Advertising	0.12	4	0.48	4	0.48	3	0.36
Market Share	0.12	4	0.48	3	0.36	2	0.24
Brand Image	0.10	4	0.40	4	0.40	3	0.30
Customer Loyalty	0.10	3	0.30	3	0.30	2	0.20
Product Quality	0.12	4	0.48	4	0.48	4	0.48
Product Range	0.08	3	0.24	4	0.32	3	0.24
Distribution	0.10	4	0.40	3	0.30	3	0.30
Price Competition	0.08	3	0.24	3	0.24	3	0.24
Geographical Expansion	0.10	4	0.40	3	0.30	2	0.20
Total Score	1.00		3.74		3.42		2.80

3.0 CAPACITY PLANNING

The production system design planning considers input requirements, conversion process and output. After considering the forecast and long-term planning, organization should undertake capacity planning.

Capacity is defined as the ability to achieve, store or produce. For an organization, capacity would be the ability of a given system to produce output within the specific time period. In operations management, capacity is referred as an amount of the input resources available to produce relative output over period of time.

In general terms, capacity is referred as maximum production size/quantity, which can be attained within a normal working schedule.

Capacity planning is the process of determining the production capacity needed by an organization to meet changing demands for its products.

3.1 Measures of Capacity

No single capacity measure is applicable to all types of situations. Hospitals measure capacity as the number of patients that can be treated per day; an airline measures capacity as available seat-miles (ASMs) per month; a theater measures capacity as number of seats; and a job shop measures capacity as number of machine hours. In general, capacity can be expressed in one of these two ways: output measures or input measures.

3.1.1 Output measures are the usual choice for facilities that produce standard products in high volumes, in a limited number of models using production processes with rigid flow patterns. Nissan Motor Company states capacity at its Tennessee plant to be 450,000 vehicles per year. The plant produces only one type of vehicle, making capacity easy to measure. However, the Navistar plant in Springfield Ohio produces vehicles that range from heavy-duty trucks to school busses. The inherent variation in production time requirements makes a single figure for capacity much less meaningful.

3.1.2 Input measures are the usual choice for facilities that produce custom products in low volumes using processes with flexible flow patterns. For example, in a photocopy shop, capacity can be measured in machine hours or number of machines. Demand, which invariably is expressed as output rate, must be converted to an input measure. Only after making the conversion can a manager compare demand requirements and capacity

on an equivalent basis. For example, the manager of a copy center must convert its annual demand for copies from different clients to the number of machines required.

3.2 Functions of Capacity Planning

The three most useful functions of capacity planning are design capacity, effective capacity and actual capacity.

3.2.1 Design capacity refers to the maximum designed service capacity or output rate. Design capacity can also be defined as the maximum amount of work that an organization is capable of completing in a given period.

3.2.2 Effective capacity is the maximum amount of work that an organization is capable of completing in a given period due to constraints such as quality problems, delays, material handling, etc. Effective capacity is the design capacity minus personal and other allowances. The maximum output that a process or firm can economically sustain under normal conditions is its effective capacity. In some organizations, effective capacity implies a one-shift operation; in others it implies a three-shift operation.

3.2.3 Actual Capacity is the maximum exact output obtained in a given period. It is sometimes referred to as average output/capacity.

These three functions of capacity can be used to find the efficiency and utilization. These are described below:

3.2.4 Utilization

Capacity planning requires knowledge of current capacity and utilization. Utilization, or the degree to which equipment, space, or labor is currently being used, is expressed as a percentage:

$$\text{Utilization} = \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100\%$$

The average (actual) output rate and the design capacity must be measured in the same terms (time, customers, units, or Naira). The utilization rate indicates the need for adding extra capacity or eliminating unneeded capacity.

$$\text{Efficiency} = \frac{\text{Actual Output}}{\text{Effective Capacity}} \times 100\%$$

Capacity planning is essential in determining optimum utilization of resource and plays an important role in decision-making process, for example, extension of existing operations, modification to product lines, starting new products, etc.

3.2.5 Peak Capacity

The maximum output that a process or facility can achieve under ideal conditions is called peak capacity. When capacity is measured relative to equipment alone, the appropriate measure is rated capacity: an engineering assessment of maximum annual output, assuming continuous operation except for an allowance for normal maintenance and repair downtime. Peak capacity can be sustained for only a short time, such as a few hours in a day or a few days in a month. A firm reaches it by using marginal methods of production, such as excess overtime, extra shifts, temporarily reduced maintenance activities, overstaffing, and subcontracting. Although they can help with temporary peaks, these options can't be sustained for long. Employees don't want to work excessive overtime for extended periods, overtime and night shift premiums drive up costs, and quality drops.

3.3 Strategic Capacity Planning

A technique used to identify and measure overall capacity of production is referred to as strategic capacity planning. Strategic capacity planning is utilized for capital intensive resource like plant, machinery, labour, etc.

Strategic capacity planning is essential as it helps the organization in meeting the future requirements of the organization. Planning ensures that operating cost is maintained at a minimum possible level without affecting the quality. It ensures the organization remain competitive and can achieve the long-term growth plan.

3.4 Capacity Planning Classification

Capacity planning based on the timeline is classified into three main categories; long term, medium term and short term.

3.4.1 Long Term Capacity: This is the strategic capacity planning undertaken by organization for period exceeding three (3) years. Long term/range capacity of an organization is dependent on various other capacities like:

design capacity, production capacity, sustainable capacity and effective capacity.

Design capacity is the maximum output possible as indicated by equipment manufacturer under ideal working condition.

Production capacity is the maximum output possible from equipment under normal working condition or day.

Sustainable capacity is the maximum production level achievable in realistic work condition and considering normal machine breakdown, maintenance, etc.

Effective capacity is the optimum production level under pre-defined job and work-schedules, normal machine breakdown, maintenance, etc.

3.4.2 Medium Term Capacity: The strategic capacity planning undertaken by organization for 2 to 3 years of a time frame is referred to as medium term capacity planning.

3.4.3 Short Term Capacity: The strategic planning undertaken by organization for a daily weekly or quarterly time frame is referred to as short term capacity planning.

3.5 Goal of Capacity Planning

The ultimate goal of capacity planning is to meet the current and future levels of requirements at a minimal wastage. The types of capacity planning based on goal are lead capacity planning, lag strategy planning, match strategy planning and adjustable strategy planning.

3.5.1 Lead strategy is adding capacity in anticipation of an increase in demand. Lead strategy is an aggressive strategy with the goal of luring customers away from the company's competitors by improving the service level and reducing lead time. It is also a strategy aimed at reducing stock-out costs. A large capacity does not necessarily imply high inventory levels, but it can imply in higher cycle stock costs. Excess capacity can also be rented to other companies.

Advantage of lead strategy:

- (i) First, it ensures that the organization has adequate capacity to meet all demand, even during periods of high growth. This is especially

important when the availability of a product or service is crucial, as in the case of emergency care or hot new product. For many new products, being late to market can mean the difference between success and failure.

- (ii) Another advantage of a lead capacity strategy is that it can be used to pre-empt competitors who might be planning to expand their own capacity. Being the first in an area to open a large grocery or home improvement store gives a retailer a definite edge.
- (iii) More so, many businesses find that overbuilding in anticipation of increased usage is cheaper and less disruptive than constantly making small increases in capacity.

Of course, a lead capacity strategy can be very risky, particularly if demand is unpredictable or technology is evolving rapidly.

3.5.2 Lag strategy refers to adding capacity only after the organization is running at full capacity or beyond due to increase in demand. This is a more conservative strategy and opposite of a lead capacity strategy. It decreases the risk of waste, but it may result in the loss of possible customers either by stock out or low service levels.

Advantage of lag strategy:

Three clear advantages of this strategy are:

- (i) A reduced risk of overbuilding.
- (ii) Greater productivity due to higher utilization levels.
- (iii) The ability to put off large investments as long as possible.

Organizations that follow this strategy often provide mature, cost-sensitive products or services.

3.5.3 Match strategy is adding capacity in small amounts in response to changing demand in the market. This is a more moderate strategy.

3.5.4 Adjustment strategy is adding or reducing capacity in small or large amounts due to consumer's demand, or, due to major changes to product or system architecture.

3.6 Factors Affecting Capacity Planning

Effective capacity planning is dependent upon factors like

- i. **Facilities:** The size and provision for expansion are key in the design of facilities. Other facility factors include locational factors (transportation

costs, distance to market, labour supply, energy sources, etc.). The layout of the work area can determine how smoothly work can be performed.

- ii. **Product and Service Factors:** The more uniform the output, the more opportunities there are for standardization of methods and materials. This leads to greater capacity.
- iii. **Process Factors:** Quantity capability is an important determinant of capacity, but so is output quality. If the quality does not meet standards, then output rate decreases because of need of inspection and rework activities. Process improvements that increase quality and productivity can result in increased capacity. Another process factor to consider is the time it takes to change over equipment settings for different products or services.
- iv. **Human Factors:** the tasks that are needed in certain jobs, the array of activities involved and the training, skill, and experience required to perform a job all affect the potential and actual output. Employee motivation, absenteeism, and labour turnover all affect the output rate as well.
- v. **Policy Factors:** Management policy can affect capacity by allowing or not allowing capacity options such as overtime or second or third shifts.
- vi. **Operational Factors:** Scheduling problems may occur when an organization has differences in equipment capabilities among different pieces of equipment or differences in job requirements. Other areas of impact on effective capacity include inventory stocking decisions, late deliveries, purchasing requirements, acceptability of purchased materials and parts, and quality inspection and control procedures.
- vii. **Supply Chain Factors:** Questions include: What impact will the changes have on suppliers, warehousing, transportation, and distributors? If capacity will be increased, will these elements of the supply chain be able to handle the increase? If capacity is to be decreased, what impact will the loss of business have on these elements of the supply chain?
- viii. **External Factors:** Minimum quality and performance standards can restrict management's options for increasing and using capacity. Inadequate planning can be a major limiting determinant of effective capacity.

3.7 Steps in the Capacity Planning Process

- (i) Estimate future capacity requirements.
- (ii) Evaluate existing capacity and facilities and identify gaps.
- (iii) Identify alternatives for meeting requirements.
- (iv) Conduct financial analyses of each alternative.
- (v) Assess key qualitative issues for each alternative.
- (vi) Select the alternative to pursue that will be best in the long term.
- (vii) Implement the selected alternative.
- (viii) Monitor results

Example 3.1: If operated around the clock under ideal conditions; the fabrication department of an engine manufacturer can make 100 engines per day. Management believes that a maximum output rate of only 45 engines per day can be sustained economically over a long period of time. Currently the department is producing an average of 50 engines per day. What is the:

- (i) utilization of the department?
- (ii) Efficiency of the department?

Solution

Actual Output = 50 engines/day;

Design Capacity = 100 engines/day;

Effective Capacity = 45 engines/day

$$\text{Utilization} = \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100\% = \frac{50}{100} \times 100\% = 50\%$$

$$\text{Efficiency} = \frac{\text{Actual Output}}{\text{Effective Capacity}} \times 100\% = \frac{50}{45} \times 100\% = 111\%$$

4.0 MATERIAL MANAGEMENT

Definition

Material may be defined as equipment, apparatus and supplies procured, stocked and utilized by an organization.

Management is a process which is applied to convert inputs into outputs (goods or services) or a process of planning, organizing, staffing, directing and controlling resources, usually by a manager.

Material Management is a scientific technique concerned with planning, organizing & control of flow of materials from their initial purchase to destination. It can also be defined as an organizational concept in which a single manager has the authority and responsibility for all activities, principally concerned with the flow of materials into an organization- purchasing, production, planning, scheduling, incoming traffic, inventory control, receiving and stores normally included.

Material Management can also be defined as the planning and control of the functions supporting the complete cycle (flow) of materials, and the associated flow of information. Also called materials planning.

4.1 AIMS OF MATERIAL MANAGEMENT

The aims of material management are to get material at:

- (i) The right quality.
- (ii) The right quantity.
- (iii) The right time.
- (iv) The right place.
- (v) The right cost.

4.2 PURPOSE OF MATERIAL MANAGEMENT

- (i) To gain economy in purchasing.
- (ii) To satisfy the demand during period of replenishment.
- (iii) To reserve stock in order to avoid stock-out.
- (iv) To stabilize fluctuations in consumption.
- (v) To provide reasonable level of client services.

4.3 OBJECTIVES OF MATERIAL MANAGEMENT

4.3.1 Primary Objectives are to achieve the following:

- (i) Right price
- (ii) High turnover
- (iii) Low procurement and storage cost

- (iv) Continuity of supply
- (v) Consistency in quality
- (vi) Good supplier relations
- (vii) Development of personnel
- (viii) Good information system

4.3.2 Secondary Objectives are:

- (i) Forecasting material requirement levels
- (ii) Inter-departmental harmony
- (iii) Product improvement
- (iv) Standardization
- (v) Make or buy decision
- (vi) Favourable reciprocal relations

4.4 FOUR BASIC NEEDS OF MATERIAL MANAGEMENT

- (i) To have adequate materials on hand when needed.
- (ii) To pay the lowest possible prices, consistent with quality and value requirements for materials.
- (iii) To minimize inventory investment.
- (iv) To operate efficiently.

4.5 FUNCTIONS OF MATERIALS MANAGEMENT

Materials management is one of the important activities of business. There is no general agreement about precisely what activities are embraced by materials management. Some managers would associate materials management with their material or production control departments, which schedule materials requirements and may also control inventories of both raw materials and in-process materials. Others would associate it with the activities of their purchasing departments in dealing with outside suppliers.

If we analyse the total cost of any product, nearly 60 to 70% is because of materials. Only the rest is for labour, overhead and profit. So any reduction in the material cost, even by a very little percentage will give rise to a greater profit. Moreover, the materials management being a staff function, the introduction of new techniques to reduce the cost of the product is much easier than in any other field.

Hence, the rate of return on capital employed is of prime concern and is given by the ratio:

$$\text{Rate of Return (ROR)} = \frac{\text{Profit}}{\text{Capital employed}}$$

$$= \frac{\text{Profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Fixed Assets} + \text{Current Assets}}$$

$$= \text{Profitability} \times \text{Capital turnover ratio}$$

So as to increase the rate of return on investment, one way is to increase the capital turnover ratio. For this, if capital employed is reduced, naturally capital turnover ratio will go high. Fixed assets constitute capital already sunk and the only scope for improving the Return on Investment (ROI) lies in the efficient management of materials which constitute the bulk of current assets.

As materials constitute the major cost component, large amount of capital is locked up in materials with the associated burden interest which further increases the cost of the product.

So, because of the greatest percentage of cost associated with materials and also any possible reduction in material cost will result in the increase of profit, the industries are now thinking of introducing the concept of scientific materials management.

4.6 SCOPE OF MATERIALS MANAGEMENT:

Materials Management strives to ensure that the material cost component of the total product cost be the least. In order to achieve this, the control is exercised in the following fields.

1. Materials Planning.
2. Purchasing.
3. Store Keeping.
4. Inventory Control.
5. Receiving, Inspection and Despatching.
6. Value Analysis, Standardization and Variety Reduction.
7. Materials Handling & Traffic.
8. Disposal of Scrap and Surplus, Material Preservation.

The function of **material planning** department is to plan for the future procurement of all the required materials as per the production schedule. At the time of material planning, the budget allocated for the materials will also be critically reviewed, for better control.

After material planning, **purchasing** is to be done. Purchasing department buys material based on the purchase requisitions from user departments and stores departments and annual production plan.

There are four basic purchasing activities.

- a) Selecting suppliers, negotiating and issuing purchase orders.
- b) Expediting delivery from suppliers.
- c) Acting as liaison between suppliers and other company departments.
- d) Looking for new products, materials, and suppliers that can contribute to company objectiveness.

At the time of purchase, right quantity and quality of materials must be purchased at right time, at the lowest possible cost through an efficient purchasing system, to derive maximum benefit. Purchasing is done based on 'make or buy' decisions and also using Programme Evaluation Review Technique (PERT) / Critical Path Method (CPM) effectively.

When the items are purchased, proper **storage** facilities must be provided so that, the wastage is reduced to a minimum. Sometimes to protect the quality, greater care must be taken during storage.

The duties of the **inventory control** department is to decide about the types of ordering system, fixing the safety stock limits, fixing up the reorder level & maximum / minimum stock level.

The responsibility of **Receiving, inspection and despatching** department is to receive the materials when delivered by the suppliers. After receiving it, the quantity and quality must be checked. Production parts and materials are checked against blueprints and specifications. Non-production items are also reviewed. When once it is as per the specifications given, the goods will be accepted.

The **Value Analysis and Standardization** offer greatest scope, in reducing the materials cost. It also reduces the number of varieties and also helps in finding the substitute for the materials at lesser cost.

Materials handling section is responsible for the transport of materials to various departments. There are four basic traffic activities.

- a) Selecting common or charter carriers and routings for despatch / shipments as required.

b) Tracing in-bound shipments of material in short supply as requested by production control or purchasing. Assisting customers in tracing out-bound shipments when asked.

c) Auditing invoices from carriers and filing claims for refunds of excess charges or for damaged shipments when required.

d) Developing techniques to reduce transportation cost. This may involve negotiation with competing shippers, special studies in selecting the most advantageous plant location for new products, analysis of tariffs, and negotiation of any number of special arrangements for handling certain traffic.

e) The activity includes packaging of finished product, labelling and loading of end products in the trades.

Finally, the disposal of scrap and surplus must be done periodically to release the capital locked in those items.

4.7 NON-PRODUCTION STORES:

Techniques and procedures used to control non-production material (office supplies, perishable tools, and maintenance, repair, and operating supplies) resemble those used for production material, although they are usually less elaborate. Specifically, the stores department:

a) Maintains physical stocks of non-production items to be drawn on as needed for operations or maintenance.

b) Manages inventories of non-production materials and prepares purchase requisitions for needed material when stocks drop to the re-order point.

c) Keeps records and maintains controls to prevent duplication of inventories, minimize losses from pilferage and spoilage and prevent stock-outs.

4.8 NEED FOR INTEGRATED CONCEPT:

In an integrated set-up, the materials manager is responsible to exercise control and coordinates with an overview that ensures proper balance of conflicting objectives of the individual functions. Integration also helps in the rapid transfer of data, through effective and formal communication channels. This is crucial as the materials management function usually involves handling a vast amount of data. Therefore, integrating the various functions ensures that message channels are shortened and the various functions identify themselves to a common materials management department which, in turn, results in greater co-ordination and better control.

4.9 ADVANTAGES IN INTEGRATED MATERIALS MANAGEMENT CONCEPT

Organizations which have gone in a big way for the integrated materials management usually enjoy the following advantages:

(i) Better Accountability:

Through centralization of authority and responsibility for all aspects of materials function, a clear cut accountability is established. This helps in evaluating the performance of materials management in an objective manner.

(ii) Better Co-Ordination:

When a central materials manager is responsible for all functions, the departments under the materials manager create an identity which is common. This results in better support and co-operation in the accomplishment of the materials function. The user departments also find that they have to approach one department for discussing and solving their materials problems. This creates an atmosphere of trust and generally better relations between the user departments and the materials management department.

(iii) Better Performance:

As all the inter-related functions are integrated organizationally, greater speed and accuracy results in improved communication. Need for materials are promptly brought to notice by materials planning. Purchase department is fed with stock levels and order status by stores and inventory control departments. All this calls for judicious decisions leading to lower costs, better inventory in paper work.

(iv) Adaptability to EDP:

The centralization of the materials function has made it possible to design data processing system. All information with regard to materials function is centralized under the integrated materials management function. This has facilitated the collection and analysis of data, leading to better decisions. Advanced and efficient electronic data processing systems can be economically introduced under integrated set-up.

(v) Miscellaneous Advantage:

Under a Centralized Materials Manager, a team spirit is inculcated and this results in better morale and co-operation. The opportunities and exposure available for the individuals for growth and development are better in an integrated set-up.

4.10 Economic Order Quantity

In stock management, Economic Order Quantity (EOQ) is an important inventory management system that demonstrates the quantity of an item to reduce the total cost of both handling of inventory (Handling Cost) and order processing (Ordering Cost). The purpose of determining the EOQ is to minimize the Total Incremental Cost (TIC), beyond the cost of purchasing of a product/material, in consideration of two main total costs: Total Ordering Cost (TOC) and Total Handling Cost (THC).

EOQ as a model has been introduced in 1913 by Ford W. Harris. With respect to an item to be ordered, from a business point of view, the EOQ model establishes the amount of quantity to be placed in an order in consideration of minimizing the annual total cost of inventory handling and order processing. In this context, these specific two types of costs are the main categories of determining the EOQ in its basic explanation. However, the model has been presented with certain assumptions for the initial understanding; and from that point onward, its extensions are used widely in businesses, especially in inventory management.

4.10.1 Assumptions of the Basic Model of Economic Order Quantity (EOQ)

The determination of EOQ consists of the following assumptions:

- a) The EOQ will be determined for every product individually in a business.
- b) Annual requirement (Demand) for product in units is known with certainty.
- c) Ordering cost is known and constant throughout the year.
- d) Inventory handling cost is known and constant throughout the year. Notably, if the handling cost of an item is given as the percentage of price of the item, the unit price of the item remains same throughout the year.
- e) No cash or quantity discount is allowed.
- f) The ordered quantity of the product is delivered at once as a single batch.
- g) Immediate replenishment of ordered quantity on time (No delay and stock shortage).

h) Constant lead time is only allowed (no fluctuation is permitted).

4.10.2 Annual Demand of the Product

The annual requirement of a product is constant and known in its measurement units and this known as annual demand of the product (D). Using various forecasting technique, it is important for a business to predict the annual demand for the specific item, for which the business need to know the EOQ. As the demand produces the primary purchasing cost of the item, the total purchasing cost is irrelevant in determining the EOQ.

4.10.3 Order Processing Cost

The Ordering Cost refers to the cost of orders to be placed for the product in consideration of order communication, allowances to purchase officers, order printing and stationery, costs of inspection, receiving the product, and transport cost, etc. Notably, these costs remain constant and unchanged for the period, irrespective the number of orders to be placed. As the ordering cost per order is constant, the relationship between the quantity ordered and number of orders to be placed is negative, i.e., higher the quantity ordered (Q) per order, lower the number of orders to be placed; and lower the quantity ordered (Q) per order, higher the number of orders to be placed. This implies the negative relationship between the quantity ordered (Q) and total cost of order processing (TOC) as in Figures 5.1.

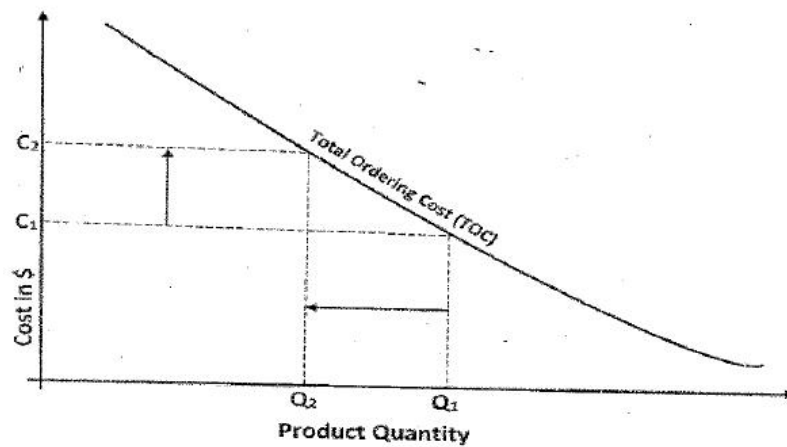


Fig. 4.1: Negative Relationship between Total Ordering Cost (TOC) and Quantity to be ordered (Q)

4.10.4 Inventory Handling Cost

This cost refers to the handling and maintaining the product inventory in workplace in consideration of warehousing costs, shrinkage loss, evaporation, deterioration and spoilage costs, insurance, warehouse rent, obsolescence, and other related overhead cost of warehouse. As these costs are constant to maintain the total demand (annual requirement) of the product, the handling cost per unit remains same. Therefore, total cost of handling has positive relationship with the number of products handled in the workplace (warehouse), i.e., higher the number of products (Q) in store, higher the total cost of handling (THC); and lower the number of products in store, lower the total handling cost (THC) as in Figure 5.2. It is notable that handling of stock would be a half the number of quantity to be ordered throughout the year; and therefore, the average stock to handle would be $Q/2$.

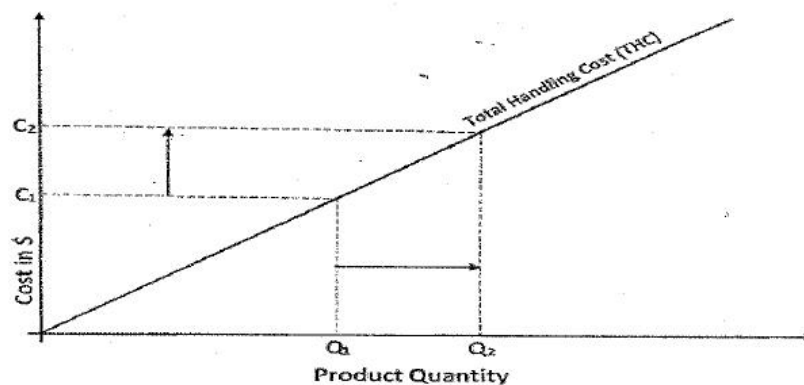


Fig. 4.2: Positive Relationship between Total Handling Cost (THC) and Quantity to be ordered (Q) for storing.

4.10.5 The Model: Economic Order Quantity (EOQ)

The basic EOQ model, with all assumptions in consideration, deal with two types of costs: Total Ordering Cost (TOC) and Total Handling (THL). It is obvious from the above explanation that these costs are moving in opposite directions, when certain number of quantities are ordered for storage purpose. Therefore, it is important for a business to find a trade-off point of ordering quantity in order to minimize the total cost of both: TOC plus THC. In this context, the quantity to be ordered to minimize the total cost of both TOC and THC is known as the Economic Order Quantity (EOQ).

As TOC has negative relationship to quantity to be ordered and THC has positive relationship to quantity to be ordered, the total minimum cost of both TOC and THC is the intersection point of both cost lines that can produce: (a) the total cost of both TOC and THC as minimum as possible; and (b) the number of quantities to be ordered (known as EOQ) to meet the minimized cost (see Figure 5.3).

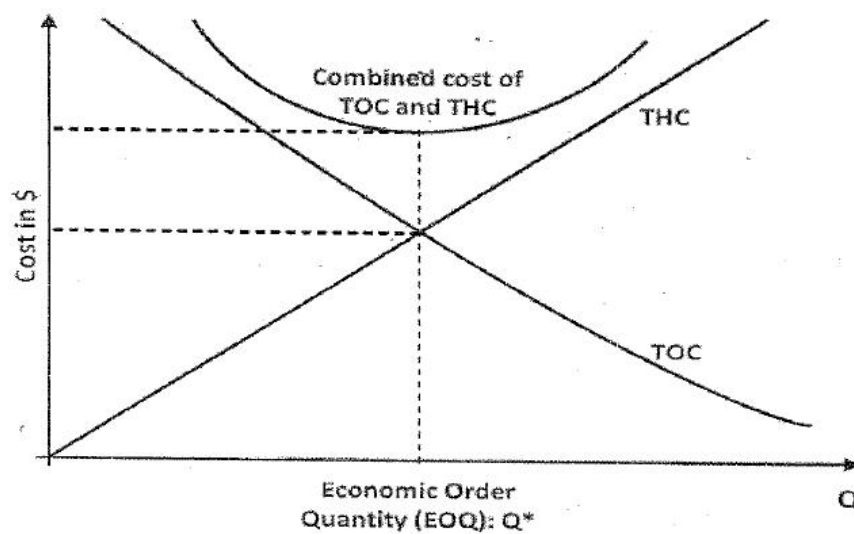


Fig. 4.3: Graphical Determination of Economic Order Quantity

Notably, from purchasing point of view, TOC and THC are the additional costs, which incur above cost of a material purchased. Therefore, the aggregation of both costs (TOC and THC) are known as Total Incremental Cost, i.e., $TIC = TOC + THC$. In the context of EOQ, TOC and THC are the additional costs incurring beyond the original purchasing cost of an item.

Generally, there are two basic methods to determine the EOQ.: (a) Trial and Error method in combination of graphical representation; and (b) Mathematical Approach – this is widely used popular method. These methods can be explained with an example for easy understanding.

Example 1: Consider a small production process, which need sawdust as raw material. The production process requires 20,000 cubic meters (m^3) annually. If an order is to be placed, every order can cost \$ 50.00 and the cost of handling one unit of cubic meter is \$ 2.

From the above, the following information is available.

Annual Demand (D) = 20000 m^3 for the sawdust,

Cost per order (CO) = \$ 50.00 and,

Handling cost per unit (CH) = \$ 2.00.

Following table can produce various possible number of cubic meters of sawdust to be ordered (this should be assumed independently) and the related cost calculations.

Table 4.1: Various number of cubic meters (quantity) and related cost calculations

Ordering Quantity (Q)	No. of Orders (N = D/Q)	Average Stock to handle (Q/2)	Total Ordering Cost (TOC) @ \$50	Total Handling Cost (THC) @ \$2	Total Incremental Cost (TIC = TOC + THC)
250	80	125	4,000.00	250.00	4,250.00
400	50	200	2,500.00	400.00	2,900.00
500	40	250	2,000.00	500.00	2,500.00
1000	20	500	1,000.00	1,000.00	2,000.00
2000	10	1000	500.00	2,000.00	2,500.00
2500	8	1250	400.00	2,500.00	2,900.00
4000	5	2000	250.00	4,000.00	4,250.00

From the table, it is possible to observe that the (last) column TIC has a minimum of \$ 2000.00, where TOC = THC = \$ 1000.00 and the quantity Q = 1000. The information available in the table can be shown in a diagram with all three costs: TIC, TOC and THC (see Figure 5.4).

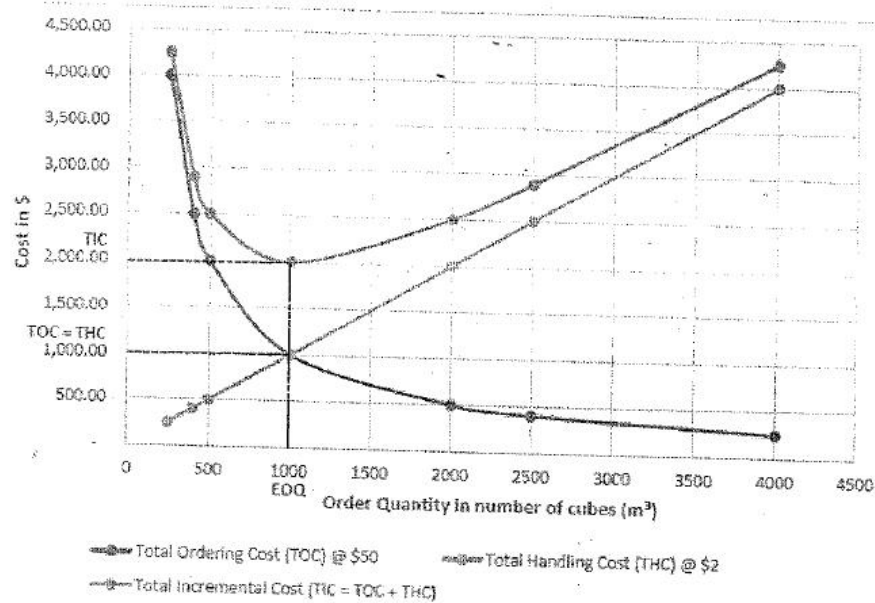


Fig. 4.4: Economic Order Quantity (EOQ) with TOC, THC and TIC

Using the same information of Example 1, the following deals with the mathematical approach, which is widely used in determining the EOQ in inventory control system. The basics of mathematical model therefore need to be illustrated to determine the EOQ to have minimum of TIC. Accordingly, it is important to determine total individual cost of both (TOC and THC), in terms of quantities to be ordered and handled.

Therefore, $TOC = \text{Cost per order} \times \frac{\text{Demand}}{\text{Quantity Ordered per year}}$

$$TOC = C_o \times \frac{D}{Q} = \frac{D \times C_o}{Q} \quad \text{and}$$

$THC = \text{Cost per unit for handling} \times$
Average quantity maintained in store for a year

$$THC = C_H \times \frac{Q}{2} = \frac{Q C_H}{2}$$

This results in $TIC = TOC + THC = \frac{D \times C_o}{Q} + \frac{Q \times C_H}{2}$

As this function TIC depends on the quantity ordered (Q) to minimize the total cost, the function needs to be differentiated with respect to Q.

This can result in: $\frac{d(TIC)}{dQ} = -\frac{D C_o}{Q^2} + \frac{C_H}{2}$

and for minimization/maximization,

$$\frac{d(TIC)}{dQ} = 0.$$

Therefore, $0 = -\frac{D C_o}{Q^2} + \frac{C_H}{2}$ and $\frac{C_H}{2} = \frac{D C_o}{Q^2}$

and solving for Q can result in

$$Q = \sqrt{\left[\frac{2 D C_o}{C_H} \right]}$$

And this must be confirmed for minimizing the TIC. To confirm the minimization of TIC for a value of Q, the second derivative of TIC should be greater than zero for the value of Q.

As such, from the first derivative, the second derivative of TIC must result in

$$\frac{d^2(TIC)}{dQ^2} = \frac{2 D C_o}{Q^3}$$

and for a value of Q, $\frac{d^2(TIC)}{dQ^2} > 0$.

Therefore, TIC optimally produce a minimum cost for the value of Q* (where Q* = EOQ).

Now, we can substitute the values available in Example 1, where Annual Demand (D) = 20000 m³ for the sawdust, Cost per order (CO) = \$ 50.00 and Handling cost per unit (CH) = \$ 2.00.

Therefore, $TOC = \frac{D \times C_o}{Q} = \frac{20,000 \times 50}{Q}$ and

$$THC = \frac{QC_H}{2} = \frac{2Q}{2} = Q$$

Then, $TIC = \frac{20,000 \times 50}{Q} + Q$

Differentiating TIC with respect to Q can result in $\frac{d(TIC)}{dQ} = -\frac{20,000 \times 50}{Q^2} + 1$

For an optimum of TIC with respect to Q, $\frac{d(TIC)}{dQ} = 0$.

i.e., $0 = -\frac{20,000 \times 50}{Q^2} + 1$ and solving for Q can result in

$$Q^2 = 20,000 \times 50 \quad \text{and} \quad Q^* = \sqrt{20,000 \times 50} = 1,000m^2$$

It is now possible to note that the number of orders to be place for the year is

$$\frac{D}{Q^*} = \frac{20,000}{1,000} = 20 \quad \text{to have the optimal TIC.}$$

$Q^* = EOQ = 1000$ can be applied to determine the TOC, THC and TIC.

$$TOC = \frac{20,000 \times 50}{1,000} = \$1000, \quad THC = \frac{2Q}{2} = \frac{2 \times 1,000}{2} = \$1000$$

And $TIC = TOC + THC = \$ (1000 + 1000) = \2000 .

Explicitly, the results are obvious about the trade off point of Q (as EOQ) between the costs of TOC and THC as equal; and the TIC results in at the minimum of the sum of costs (TOC + THC).

4.10.6 Determining TIC with Demand (D), per Order Cost (C_o) and per unit Product Handling Cost (C_H)

$$\text{As TIC} = \text{TOC} + \text{THC} = \frac{D \times C_o}{Q} + \frac{Q C_H}{2}$$

$$\text{And EOQ} = Q^* = \sqrt{\frac{2DC_o}{C_H}}$$

Now Quantity (Q) in the TIC function can be represented with $\text{EOQ} = Q^*$.

Therefore, Q can be substituted with $\sqrt{\frac{2DC_o}{C_H}}$. This can result in

$$\text{TIC} = \frac{D \times C_o}{Q^*} + \frac{Q^* C_H}{2} = \left(\frac{DC_o}{\sqrt{\frac{2DC_o}{C_H}}} + \frac{\sqrt{\frac{2DC_o}{C_H}} \times C_H}{2} \right) = \left(\frac{DC_o}{\sqrt{\frac{2DC_o}{C_H}}} + \frac{\sqrt{2DC_o C_H^2}}{2} \right)$$

$$\text{TIC} = \left(\frac{D^2 C_o^2 C_H}{2DC_o} + \sqrt{\frac{2DC_o C_H^2}{4DC_H}} \right) = \left(\sqrt{\frac{DC_o C_H}{2}} + \sqrt{\frac{DC_o C_H}{2}} \right)$$

$$\text{TIC} = \left(2 \times \sqrt{\frac{DC_o C_H}{2}} \right) = \sqrt{\frac{4DC_o C_H}{2}} = \sqrt{2DC_o C_H}$$

As in Example 1, Annual Demand (D) = 20000 m^3 for the sawdust, Cost per order (C_o) = \$ 50.00 and Handling cost per unit (C_H) = \$ 2.00,

$$\text{TIC} = \sqrt{2DC_o C_H} = \sqrt{2 \times 20,000 \times 50 \times 2} = \$ 2000.$$

4.10.7 Transforming EOQ into Economic (Optimum) Number of Orders (ENO)

When EOQ (Q^*) is determined and termed as optimal quantity to minimize the TIC, the number of orders for the year (N) can be determined as

N = Demand (D) divided by EOQ (i.e., $N = D/Q^*$).

This in other term can be interpreted that the Economic (optimal) Number of Orders, ($ENO = N^*$) can provide the minimum TIC. In this context, Q can be substituted in terms of N and the mathematical approach can be extended to determine the Economic Number of Orders (ENO) and TIC relatively, as shown below.

Therefore, $TOC = \text{Cost per order} \times \frac{\text{Demand}}{\text{Quantity Ordered per year}}$

$$TOC = C_o \times \frac{D}{Q} = \frac{DC_o}{Q} \quad \text{and as } N = \frac{D}{Q}$$

$$TOC = N \times C_o$$

$THC = \text{Cost per unit for handling} \times \text{Average quantity maintained in store for a year}$

$$THC = C_H \times \frac{Q}{2} = \frac{QC_H}{2} \quad \text{and as } N = \frac{D}{Q}$$

$$THC = \frac{C_H Q}{2} = \frac{DC_H}{2N}$$

This results in $TIC = TOC + THC = (N C_o + \frac{DC_H}{2N})$

As this function TIC depends on the quantity ordered (Q) to minimize the total cost, the function needs to be differentiated with respect to N . This can result in:

$$\frac{d(TIC)}{dN} = C_o + \frac{-DC_H}{2N^2} \quad \text{and for minimization/maximization } \frac{d(TIC)}{dQ} = 0.$$

$$\text{Therefore, } 0 = C_o + \frac{-DC_H}{2N^2} \quad \text{and} \quad C_o = \frac{DC_H}{2N^2}$$

And solving for Q can result in $N = \sqrt{\frac{DC_H}{2C_o}}$ and this must be confirmed for minimizing the TIC.

To confirm the minimization of TIC for a value of N, the second derivative of TIC should be greater than zero for the value of N.

As such, from the first derivative, the second derivative of TIC must result in

$$\frac{d^2(TIC)}{dN^2} = \frac{DC_H}{N^3} \quad \text{and for a value of } Q, \frac{d^2(TIC)}{dN^2} > 0.$$

Therefore, TIC optimally produce a minimum cost for the value of N^* (where $N^* = ENO$).

As in Example 1, Annual Demand (D) = 20,000 m^3 for the sawdust, Cost per order (C_o) = \$ 50.00 and Handling cost per unit (CH) = \$ 2.00, and substituting these values in

$$N^* = \sqrt{\frac{DC_H}{2C_o}} = \sqrt{\frac{20,000 \times 2}{2 \times 50}} = 20.$$

With respect to the number of orders to be placed, ENO can be found can be represented in a diagram with TOC, THC and TIC, respectively (see Table 4.2 and Figure 4.5):

Table 4.2: Various number of orders and related cost calculations

No. of Orders (N)	Ordering Quantity (Q = D/N)	Average Stock to handle (Q/2)	Total Ordering Cost (TOC) @ \$50	Total Handling Cost (THC) @ \$2	Total Incremental Cost (TIC = TOC + THC)
80	250	125	4,000.00	250.00	4,250.00
50	400	200	2,500.00	400.00	2,900.00
40	500	250	2,000.00	500.00	2,500.00
20	1000	500	1,000.00	1,000.00	2,000.00
10	2000	1000	500.00	2,000.00	2,500.00
8	2500	1250	400.00	2,500.00	2,900.00
5	4000	2000	250.00	4,000.00	4,250.00

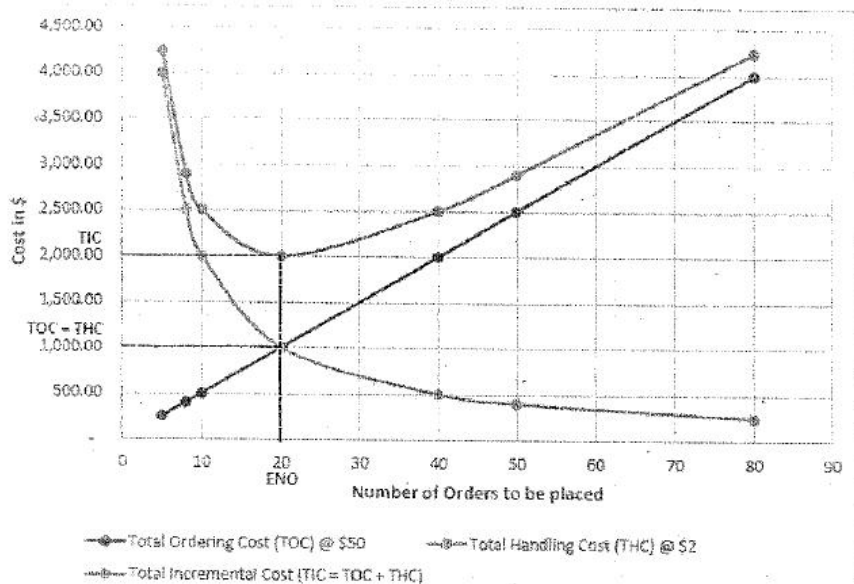


Fig. 4.5: Economic Number of Orders (ENO) with TOC, THC and TIC

It is now possible to note that the number of quantities (known as EOQ in other term) to be placed in an order is $(D/N^*) = (20000/20) = 1000$, to have the optimal TIC.

4.10.8 Length of inventory cycle (provided with daily usage 'd' of the product)

Length of inventory cycle is a measure that gives a time period how long a batch of EOQ can last in the storage. As production/supply of the items takes place, the daily usage (d) of the item becomes reduction in the EOQ stored. Therefore, length of inventory cycle indicated how many times (T) of daily usage of the item in total equates the EOQ.

Therefore, $EOQ = T \times d$ where T = length of inventory cycle and
d = daily usage of the item.

$$T = \left(\frac{EOQ}{d}\right) = \left(\frac{Q^*}{d}\right)$$

Note that the daily usage should have been provided with certainty (consider it as an assumption). Referring to the information above, EOQ = 1000 units and assume daily use (d) of the item is 100 units. Therefore, length of inventory cycle

$$T = \left(\frac{EOQ}{d}\right) = \frac{1000}{100} = 10 \text{ days}$$

4.10.9 Determining number of annual working days

It is also notable that every batch of EOQ last for certain period (10 days in the example above) and this will happen for every order of EOQ in a year. Therefore, the T (=10 days) time cycle is applicable for every order. As EOQ can be transformed into Economic Number of Orders ($ENO = N^*$) in a year and every order (a batch of EOQ) last for T (=10 days) time as a cycle, the annual number of working days in a year on this product can be determined as:

Annual number of working days = $N^* \cdot T^* = (20) \times (10) = 200$ days

Note in the example, $N^* = (D/EOQ) = (20000/1000) = 20$ and

$T^* = (EOQ/d) = (1000/100) = 10$.

5.0 SCHEDULING

OPERATIONS SCHEDULING

Scheduling pertains to establishing both the timing and use of resources within an organization. Under the operations function (both manufacturing and services), scheduling relates to use of equipment and facilities, the scheduling of human activities, and receipt of materials.

While issues relating to facility location, plant and equipment acquisition are considered long term and aggregate planning is considered intermediate term, operations scheduling is considered to be a short-term issue. As such, in the decision-making hierarchy, scheduling is usually the final step in the transformation process before the actual output (e.g., finished goods) is produced. Consequently, scheduling decisions are made within the constraints established by these longer-term decisions. Generally, scheduling objectives deals with trade-offs among conflicting goals for efficient utilization of labour and equipment, lead time, inventory levels, and processing times.

Byron Finch notes that effective scheduling has recently increased in importance. This increase is due in part to the popularity of lean manufacturing and just-in-time. The resulting drop in inventory levels and subsequent increased replenishment frequency has greatly increased the probability of the occurrence of stock-outs. In addition, the Internet has increased pressure to schedule effectively. "Business to customer" (B2C) and "business to business" (B2B) relationships have drastically reduced the time needed to compare prices, check product availability, make the purchase, etc. Such instantaneous transactions have increased the expectations of customers, thereby, making effective scheduling a key to customer satisfaction. It is noteworthy that there are over 100 software scheduling packages that can perform schedule evaluation, schedule generation, and automated scheduling. However, their results can often be improved through a human scheduler's judgment and experience.

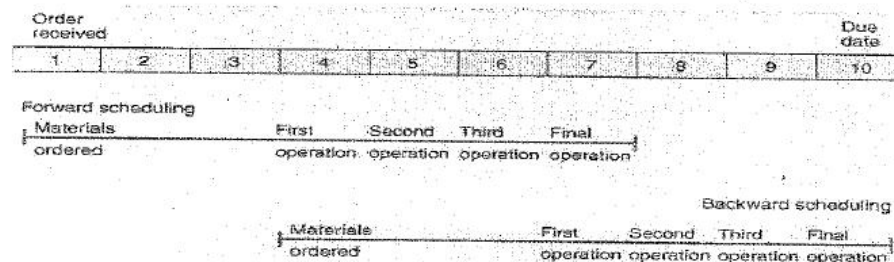
Scheduling is the process of arranging, controlling and optimizing work and workloads in a production process.

Key Words Definition:

- **Routing:** The operations to be performed, their sequence, the work centers, & the time standards.
- **Bottleneck:** A resource whose capacity is less than the demand placed on it.
- **Due date:** When the job is supposed to be finished.
- **Slack:** The time that a job can be delayed & still finish by its due date.
- **Queue:** A waiting line.

There are two general approaches to scheduling: forward scheduling and backward scheduling.

- **Forward Scheduling** – starts processing when a job is received.
- **Backward Scheduling** – begin scheduling the job's last activity so that the job is finished on due date.



Companies use backward and forward scheduling to allocate plant and machinery resources, plan human resources, plan production processes and purchase materials. As long as the concepts are applied properly, the choice of methods is not significant.

With forward scheduling, the scheduler selects a planned order release date and schedules all activities from this point forward in time. Forward scheduling is planning the tasks from the date resources become available to determine the shipping date or the due date.

With backward scheduling, the scheduler begins with a planned receipt date or due date and moves backward in time, according to the required processing times, until he or she reaches the point where the order will be released. Backward scheduling is planning the tasks from the due date or required-by date to determine the start date and/or any changes in capacity required.

The benefits of production scheduling include:

- Process change-over reduction
- Inventory reduction, levelling
- Reduced scheduling effort
- Increased production efficiency
- Labour load levelling
- Accurate delivery date quotes
- Real time information

Factors that Influence Scheduling include:

- due dates or shipping dates.
- the types of jobs to be processed and the different resources that can process each
- process routings
- processing times
- setup times
- changeover times
- resource availability
- number of shifts
- downtime, and
- planned maintenance.

LOADING

Loading involves assigning jobs to work centres and to various machines in the work centres. If a job can be processed on only one machine, no difficulty is presented. However, if a job can be loaded on multiple work centres or machines, and there are multiple jobs to process, the assignment process becomes more complicated. The scheduler needs some way to assign jobs to the centres in such a way that processing and setups are minimized along with idle time and throughput time.

Two approaches are used for loading work centres:

- **Infinite loading:** Ignores capacity constraints, but helps identify bottlenecks in a proposed schedule to enable proactive management
- **Finite loading:** Allows only as much work to be assigned as can be done with available capacity – but doesn't prepare for inevitable slippage

With infinite loading jobs are assigned to work centres without regard for capacity of the work centre. Priority rules are appropriate for use under the infinite loading approach. Jobs are loaded at work centres according to the chosen priority rule. This is known as **vertical loading**.

In finite loading projects the actual start and stop times of each job at each work centre. Finite loading considers the capacity of each work centre and compares the processing time so that process time does not exceed capacity. With finite loading the scheduler loads the job that has the highest priority on all work centres it will require. Then the job with the next highest priority is loaded on all required work centres, and so on. This process is referred to as **horizontal loading**. The scheduler using finite loading can then project the number of hours each work centre will operate. A drawback of horizontal loading is that jobs may be kept waiting at a work centre, even though the work centre is idle. This happens when a higher priority job is

expected to arrive shortly. The work centre is kept idle so that it will be ready to process the higher priority job as soon as it arrives.

With vertical loading the work centre would be fully loaded. Of course, this would mean that a higher priority job would then have to wait to be processed since the work centre was already busy. The scheduler will have to weigh the relative costs of keeping higher priority jobs waiting, the cost of idle work centres, the number of jobs and work centres, and the potential for disruptions, new jobs and cancellations.

If the firm has limited capacity (e.g., already running three shifts), finite loading would be appropriate since it reflects an upper limit on capacity. If infinite loading is used, capacity may have to be increased through overtime, subcontracting, or expansion, or work may have to be shifted to other periods or machines.

SEQUENCING

Sequencing is concerned with determining the order in which jobs are processed. Not only must the order be determined for processing jobs at work centres but also for work processed at individual work stations. When work centres are heavily loaded and lengthy jobs are involved, the situation can become complicated. The order of processing can be crucial when it comes to the cost of waiting to be processed and the cost of idle time at work centres.

There are a number of **priority rules** or heuristics that can be used to select the order of jobs waiting for processing.

- i. **Random (R):** Pick any job in the queue with equal probability. This rule is often used as a benchmark for other rules.
- ii. **First come/first served (FC/FS):** This rule is sometimes deemed to be fair since jobs are processed in the order in which they arrive.
- iii. **Last come, first served (LCFS).**
- iv. **Shortest processing time (SPT):** The job with the shortest processing time requirement goes first. This rule tends to reduce work-in-process inventory, average throughput time, and average job lateness.
- v. **Longest processing time (LPT).**
- vi. **Earliest due date (EDD):** The job with the earliest due date goes first. This seems to work well if the firm performance is judged by job lateness.
- vii. **Critical ratio (CR):** To use this rule one must calculate a priority index using the formula $(\text{due date} - \text{now}) / (\text{lead time remaining})$. This rule is widely used in practice. The critical ratio (CR) is calculated by dividing the time remaining until a job's due date by the total shop time remaining for the job, $(\text{Time until due date}) / (\text{processing time})$, which is defined as the setup, processing, move, and expected waiting times of all remaining operations, including the operation being scheduled.

- viii. **Least work remaining (LWR):** An extension of SPT, this rule dictates that work be scheduled according to the processing time remaining before the job is considered to be complete. The less work remaining in a job, the earlier it is in the production schedule.
- ix. **Fewest operations remaining (FOR):** This rule is another variant of SPT; it sequences jobs based on the number of successive operations remaining until the job is considered complete. The fewer operations that remain, the earlier the job is scheduled.
- x. **Slack time (ST):** This rule is a variant of EDD; it utilizes a variable known as slack. Slack is computed by subtracting the sum of setup and processing times from the time remaining until the job's due date. Jobs are run in order of the smallest amount of slack.
- xi. **Slack time per remaining operation (ST/RO):** This is a variant of ST. The slack time is divided by the number of operations remaining until the job is complete with the smallest values being scheduled first (Slack) / (number of remaining operations).
- xii. **Next queue (NQ):** NQ is based on machine utilization. The idea is to consider queues (waiting lines) at each of the succeeding work centres at which the jobs will go. One then selects the job for processing that is going to the smallest queue, measured either in hours or jobs. Work is scheduled for the machining centre with the shortest queue.
- xiii. **Least setup (LSU):** This rule maximizes utilization. The process calls for scheduling first the job that minimizes changeover time on a given machine.

These rules assume that setup time and setup cost are independent of the processing sequence. However, this is not always the case. Jobs that require similar setups can reduce setup times if sequenced back to back. In addition to this assumption, the priority rules also assume that setup time and processing times are deterministic and not variable, there will be no interruptions in processing, the set of jobs is known, no new jobs arrive after processing begins, and no jobs are cancelled. While little of this is true in practice, it does make the scheduling problem manageable.

Example Using SPT and EDD at Jill's Machine Shop-Work Center 101				
	Job Time	Days to	SPT Rule	EDD Rule
Job Number	(includes Setup & Run Time)	Due Date	Sequence	Sequence
AZK111	3 days	3	EZE101	AZK111
BRU872	2 days	6	BRU872	EZE101
CUF373	5 days	8	AZK111	DBR664
DBR664	4 days	5	DBR664	BRU872
EZE101	1 day	4	FID448	CUF373
FID448	4 days	9	CUF373	FID448

How to Use Priority Rules:

1. Decide which priority rule to use
2. List all jobs waiting to be processed with their job time
3. Using priority rule determine which job has highest priority then second, third and so on

SCHEDULING SERVICE OPERATIONS

The scheduling of services often encounters problems not seen in manufacturing. Much of this is due to the nature of service, i.e., the intangibility of services and the inability to inventory or store services and the fact that demand for services is usually random. Random demand makes the scheduling of labour extremely difficult as seen in restaurants, movie theatres, and amusement parks. Since customers don't like to wait, labour must be scheduled so that customer wait is minimized. This sometimes requires the use of queuing theory or waiting line theory. Queuing theory uses estimate arrival rates and service rates to calculate an optimum staffing plan. In addition, flexibility can often be built into the service operation through the use of casual labour, on-call employees, and cross-training.

Scheduling of services can also be complicated when it is necessary to coordinate and schedule more than one resource. For example, when hospitals schedule surgery, not only is the scheduling of surgeons involved but also the scheduling of operating room facilities, support staff, and special equipment. Along with the scheduling of classes, universities must also schedule faculty, classrooms, labs, audio-visual and computer equipment, and students. To further complicate matters, cancellations are also common and can add further disruption and confusion to the scheduling process.

Instead of scheduling labour, service firms frequently try to facilitate their service operations by scheduling demand. This is done through the use of appointment systems and reservations.

Performance Measures

Flow time is the time a job spends in the service or manufacturing system.

Past due (tardiness) is the amount of time by which a job missed its due date. In this regard, a job is the object receiving service or being manufactured.

Makespan: This is the total amount of time required to complete a group of jobs. Minimizing makespan supports the competitive priorities of cost (lower inventory) and time (delivery speed).

Total Inventory: This performance measure is used to measure the effectiveness of schedules for manufacturing processes. The sum of scheduled receipts and on-hand inventories is the total inventory. Minimizing total inventory supports the competitive priority of cost (inventory holding costs).

Utilization: The degree to which equipment, space, or the workforce is currently being used, measured as the ratio of the average output rate to maximum capacity. Maximizing the utilization of a process supports the competitive priority of cost (slack capacity). These performance measures often are interrelated. For example, minimizing the average flow time tends to increase utilization. Minimizing the makespan for a group of jobs tends to increase utilization. Understanding how flow time, makespan, past due, and utilization interact can make the selection of good schedules easier.

Sequencing Jobs Operations schedules are short-term plans designed to implement the sales and operations plan. Often, several jobs must be processed at one or more workstations. Typically, a variety of tasks can be performed at each workstation. If schedules are not carefully planned to avoid bottlenecks, waiting lines may develop.

Total inventory = Scheduled receipts for all items + On-hand inventories of all items

Makespan = Time of completion of last job - Starting time of the first job

- **Job flow time:** Time a job is completed minus the time the job was first available for processing.
- Average flow time measures responsiveness.
- Average jobs in system measures amount of work-in-progress.
- **Makespan:** The time it takes to finish a batch of jobs; measure of efficiency

- Job lateness: Whether the job is completed ahead of, on, or behind schedule;
- Job tardiness: How long after the due date a job was completed, measures due date performance

Scheduling performance Calculations

Job A finishes on day 10	Job B finishes on day 13	Job C finishes on day 17	Job D ends on day 20
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Assume that all the jobs started on day 1; calculate mean flow time, average number of jobs in the system, and makespan.

- **Calculating mean flow time:**
 - $MFT = (\text{sum job flow times}) / \# \text{ of jobs}$
 $= (10+13+17+20)/4 = 60/4 = 15 \text{ days}$
- **Calculating average number of jobs in the system:**
 - $\text{Average \# Jobs} = (\text{sum job flow times}) / \# \text{ days to complete batch}$
 $= (60)/20 = 3 \text{ job}$
- **Makespan is the length of time to complete a batch**
 - $\text{Makespan} = \text{Completion time for Job D minus start time for Job A}$
 $= 20 - 0 = 20 \text{ days}$

Lateness and Tardiness are both measures related to customer service

Average tardiness is a more relevant **Customer Service** measurement as illustrated below

Example 15-5 Calculating job lateness and job tardiness				
	Completion			
Job	Date	Due Date	Lateness	Tardiness
A	10	15	-5	0
B	13	15	-2	0
C	17	10	7	7
D	20	20	0	0
		Average	0	1.75

The Complexity of Scheduling a Manufacturing Process

When a job order is received for a part, the raw materials are collected and the batch is moved to its first operation. At each workstation, the next job to

process is a decision because the arrival rate of jobs at a workstation often differs from the processing rate of the jobs at a workstation, thereby creating a waiting line. In addition, new jobs can enter the process at any time, thereby creating a dynamic environment. Such complexity puts pressure on managers to develop scheduling procedures that will handle the workload efficiently. In this section, we focus on scheduling approaches used in two environments:

- (1) Divergent flow processes and (2) line flow processes.

A manufacturer's operation with divergent flows is often called a job shop, which specializes in low- to medium-volume production and utilizes job or batch processes. The front office would be the equivalent for a service provider. Jobs in divergent flow processes are difficult to schedule because of the variability in job routings and the continual introduction of new jobs to be processed.

A manufacturer's operation with line flows is often called a flow shop, which specializes in medium- to high-volume production and utilizes line or continuous flow processes. The back office would be the equivalent for a service provider. Tasks are easier to schedule because the jobs have a common flow pattern through the system. Nonetheless, scheduling mistakes can be costly in either situation.

Sequencing Jobs Through Two Work Centers –Johnson's Rule

Johnson's Rule – a technique for minimizing makespan in a two-stage, unidirectional process

Step 1 – List the jobs and the processing time for each activity

Step 2 – Find the shortest activity processing time among the jobs not yet scheduled

- i. If the shortest Processing time is for a 1st activity, schedule that job in the earliest available position in the job sequence
- ii. If the shortest processing time is for 2nd activity, schedule that job in the last available position in the job sequence
- iii. When you schedule a job eliminate it from further consideration

Step 3 – Repeat step 2 until you have put all activities for the job in the schedule

Johnson's Rule Example: Vicki's Office Cleaners does the annual major cleaning of university buildings. The job requires mopping (1st activity) and waxing (2nd activity) of each building. Vicki wants to minimize the time it takes

her crews to finish cleaning (minimize makespan) the five buildings. She needs to finish in 20 days.

	Activity 1	Activity 2	Johnson's	Activity 1	Activity 2
Hall	Mopping (days)	Waxing (days)	Sequence	Mopping (days)	Waxing (days)
Adams Hall	1	2	Adams Hall (A)	1	2
Bryce Building	3	5	Chemistry Building (C)	2	4
Chemistry Building	2	4	Bryce Building (B)	3	5
Drake Union	5	4	Drake Union (D)	5	4
Evans Center	4	2	Evans Center (E)	4	2

Activity	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Mopping	A	C	C	B	B	B	D	D	D	D	D	E	E	E	E			
Waxing		A	A	C	C	C	C	B	B	B	B	B	D	D	D	D	E	E

SCHEDULING BOTTLENECKS

In the 1970's Eli Goldratt introduced Optimized Production Technology (OPT). OPT focused on bottlenecks for scheduling & capacity planning.

OPT PRINCIPLES

- Balance the process rather than the flow
- Non-bottleneck usage is driven by some other constraint in the system
- Usage and activation of a resource are not the same
- An hour lost at a bottleneck is lost forever, but an hour lost at a non-bottleneck is a mirage
- Bottleneck determine throughput and inventory in system
- The transfer batch does not need to be equal to the process batch
- The process batch should be variable
- Consider all constraints simultaneously. Lead times are the result of the schedule and are not predetermined.

THEORY OF CONSTRAINTS

- TOC is an extension of OPT – theory is that a system's output is determined by its constraints
 - i. Identify the bottleneck(s) in the process

- ii. Exploit (fully utilize) the bottleneck(s)
- iii. Subordinate all other decisions to Step 2 - Schedule non-bottlenecks to support maximum use of bottleneck activities
- iv. Elevate the Bottleneck(s)
- v. Do not let inertia set in

SCHEDULING FOR SERVICE ORGANIZATIONS

- **Demand management:**
 - i. Appointments & reservations
 - ii. Posted schedules
 - iii. Delayed services or backlogs (queues)
- **Scheduling Employees:**
 - i. Staff for peak demand (if cost isn't prohibitive)
 - ii. Floating employees or employees on call
 - iii. Temporary, seasonal, or part-time employees

6.0 LAYOUT PLANNING

Layout planning is deciding on the best physical arrangement of all resources that consume space within a facility. These resources might include a desk, a work centre, a cabinet, a person, an entire office, or even a department. Decisions about the arrangement of resources in a business are not made only when a new facility is being designed; they are made any time there is a change in the arrangement of resources, such as a new worker being added, a machine being moved, or a change in procedure being implemented. Also, layout planning is performed any time there is an expansion in the facility or a space reduction.

Plant layout is a plan of an optimum arrangement of facilities including personnel, operating equipment, storage space, material handling equipment and all other supporting services along with the design of the best structure to contain all these facilities.

Levels of plant layout design

There are four levels of plant layout design.

1. Site layout: shows how the building should be located in a proper way.
2. Block layout: shows the sizes of departments in the buildings.
3. Detailed layout: shows the arrangements of equipment and workstations in the departments.
4. Workstation layout: shows the locations of every part of the workstation.

Layout Planning Criteria:

The primary criteria for evaluating any layout will be the MINIMIZATION of material handling costs.

MH cost components includes;

- (i) depreciation of MH equipment,
- (ii) variable operating costs, and
- (iii) labour expenses.

MH costs are typically directly proportional to:

- (a) the frequency of movement of material, and
- (b) The length over which material is moved.

Advantages of these criteria (reduced material movements):

1. Reduction of Aisle space required.
2. Lower Work-In-Progress(WIP) levels

3. Lower throughput times
4. Less product damage and lower obsolescence
5. Reduced storage space
6. Simplified material control and scheduling
7. Less congestion in system.

Consider the following:

Location of next operation	Material Handled per move
adjacent machine	single part
across the aisle	unit load
across the plant	lot size of over 1 hour of production
another plant	one day's production

At each stage, the work-in-progress (WIP) is increasing by as much as 10 times.

6.1 OBJECTIVES OF LAYOUT PLANNING:

The objectives of a good layout are as follows:

- (i) Should provide overall satisfaction to all concerned.
- (ii) Material handling and internal transportation from one operation to the next is minimized and efficiently controlled.
- (iii) The production bottle necks and points of congestions are to be eliminated so that input raw materials and semi-finished parts move fast from one work station to another.
- (iv) Should provide high work in process turnover.
- (v) Should utilize the space most effectively; may be cubical utilization.
- (vi) Should provide worker's convenience, promote job satisfaction and safety for them.
- (vii) Should avoid unnecessary investment of capital.

- (viii) Should help in effective utilization of labour.
- (ix) Should lead to increased productivity and better quality of the product with reduced capital cost.
- (x) Should provide easy supervision.
- (xi) Should provide space for future expansion of the plant.
- (xii) Should provide proper lighting and ventilation of the areas of work stations.

6.2 PRINCIPLES OF PLANT LAYOUT:

1. Principle of Overall Integration:

According to this principle the best layout is one which provides integration of production facilities like men, machinery, raw materials, supporting activities and any other such factors which result in the best compromise.

2. Principle of Minimum Distance:

According to this principle, the movements of men and materials should be minimized.

3. Principle of Flow:

The best layout is one which arranges the work station for each separate process in same order or sequence that assembles the materials.

4. Principle of Cubic Space Utilization:

According to this, the best layout utilizes cubic space i.e. space available both in vertical and horizontal directions is most economically and effectively utilized.

5. Principle of Safety, Security and Satisfaction:

According to this principle, best layout is one which provides satisfaction and safety to all workers and safeguards the plant and machinery against fire, theft, etc.

6. Principle of Flexibility:

In automotive and other allied industries where models of products change after sometime, the principle of flexibility provides adoption and rearrangements at a minimum cost and least inconvenience.

6.3 TYPES OF LAYOUT

Keeping in view the type of industry and volume of production, the type of layout to be selected is to be decided from the following:

1. Product or Line Layout
2. Process or Functional Layout.
3. Fixed Position Layout.
4. Combination type of Layout.

1. Product or Line Layout:

If all the processing equipment and machines are arranged according to the sequence of operations of the product, the layout is called product type of layout. In this type of layout, only one product or one type of products is produced in an operating area. This product must be standardized and produced in large quantities in order to justify the product layout.

The raw material is supplied at one end of the line and goes from one operation to the next quite rapidly with a minimum work in process, storage and material handling.

Advantages offered by Product Layout:

- (i) Lowers total material handling cost.
- (ii) There is less work in processes.
- (iii) Better utilization of men and machines.
- (iv) Less floor area is occupied by material in transit and for temporary storages.
- (v) Greater simplicity of production control.
- (vi) Total production time is also minimized.

Limitations of Product Layout:

- (i) No flexibility which is generally required is obtained in this layout.

- (ii) The manufacturing cost increases with a fall in volume of production.
- (iii) If one or two lines are running light, there is a considerable machine idleness.
- (iv) A single machine break down may shut down the whole production line.
- (v) Specialized and strict supervision is essential to avoid downtime.

2. Process or Functional Layout:

The process layout is particularly useful where low volume of production is needed. If the products are not standardized, the process layout is more desirable, because it has greater process flexibility than others. In this type of layout, the machines are not arranged according to the sequence of operations but are arranged according to the nature or type of the operations. This layout is commonly suitable for non-repetitive jobs.

Same type of operation facilities are grouped together such as lathes will be placed at one place, all the drill machines are at another place and so on. Therefore, the process carried out in that area is according to the machine available in that area.

Advantages of Process Layout:

- (i) There will be less duplication of machines. Thus, total investment in equipment purchase will be reduced.
- (ii) It offers better and more efficient supervision through specialization at various levels.
- (iii) There is a greater flexibility in equipment and man power thus load distribution is easily controlled.
- (iv) Better utilization of equipment available is possible.
- (v) Break down of equipment can be easily handled by transferring work to another machine/work station.
- (vi) There will be better control of complicated or precision processes, especially where much inspection is required.

Limitations of Process Layout:

- (i) There are long material flow lines and hence the expensive handling is required.
- (ii) Total production cycle time is more owing to long distances and waiting at various points.
- (iii) Since more work is in queue and waiting for further operation hence bottle necks occur.
- (iv) Generally, more floor area is required.
- (v) Since work does not flow through definite lines, counting and scheduling is more tedious.
- (vi) Specialization creates monotony and there will be difficulty for the laid-off workers to find job in other industries.

3. Fixed Position Layout:

This type of layout is the least important for today's manufacturing industries. In this type of layout, the major component remains in a fixed location, other materials, parts, tools, machinery, man power and other supporting equipment's are brought to this location.

The major component or body of the product remain in a fixed position because it is too heavy or too big and as such it is economical and convenient to bring the necessary tools and equipment to the work place along with the man power. This type of layout is used in the manufacture of boilers, hydraulic machines and steam turbines and ships etc.

Advantages Offered by Fixed Position Layout:

- (i) In-process material movement is reduced
- (ii) Capital investment is minimized.
- (iii) The task is usually done by gang of operators, hence continuity of operations is ensured
- (iv) Production centers are independent of each other. Hence, effective planning and loading can be made. Thus total production cost will be reduced.

(v) It offers greater flexibility and allows change in product design, product mix and production volume.

Limitations of Fixed Position Layout:

- (i) Highly skilled man power is required.
- (ii) Movement of machines equipment's to production centre may be time consuming.
- (iii) Complicated fixtures may be required for positioning of jobs and tools. This may increase the cost of production.

4. Combination Type of Layout:

Nowadays in pure state any one form of layouts discussed above is rarely found. Therefore, generally the layouts used in industries are the compromise of the above mentioned layouts. Every layout has got certain advantages and limitations. Therefore, industries would like to use any type of layout as such.

Flexibility is a very important factor, so layout should be such which can be modelled according to the requirements of industry, without much investment. If the good features of all types of layouts are connected, a compromise solution can be obtained which will be more economical and flexible.

6.4 LAYOUT PLANNING TOOLS

6.4.1 Systematic Layout Planning

The **systematic layout planning** (SLP) is a tool used to arrange a workplace in a plant by locating areas with high frequency and logical relationships close to each other. The process permits the quickest material flow in processing the product at the lowest cost and least amount of handling.

6.4.2 Group or Cellular Layout

- In cellular manufacturing, production work stations and equipment are arranged in a sequence that supports a smooth flow of materials and components through the production process with **Minimal Transport or Delay**.

- Implementation of this Lean Method often represents an attempt to combine the flexibility of a process layout with the efficiency of a product layout.
- Rather than processing multiple parts before sending them on to the next machine or process step (as is the case in batch-and-queue, or large-lot production), cellular manufacturing aims to move products through the manufacturing process one-piece at a time, at a rate determined by customers' needs.
- The approach seeks to minimize the time it takes for a single product to flow through the entire production process.
- This One-Piece Flow method includes specific analytical techniques for assessing current operations and designing a new cell-based manufacturing layout that will shorten cycle times and changeover times.
- To make the cellular design work, an organization must often replace large, high volume production machines with small, flexible, "right-sized" machines to fit well in the cell.
- Equipment often must be modified to stop and signal when a cycle is complete or when problems occur, using a technique called Autonomation (or Jidoka).
- This transformation often shifts worker responsibilities from watching a single machine, to managing multiple machines in a production cell. **(One-worker, multiple-machines) (OWMM)**
- Cellular manufacturing can also provide companies with the flexibility to vary product type or features on the production line in response to specific customer demands.
- Using this technique, production capacity can be incrementally increased or decreased by adding or removing production cells.

Characteristics

- Limited number of parts or models are processed with line flows.
- Based on the concept of group technology (GT), dissimilar machines are grouped into work centers, called cells, to process parts with similar shapes or processing requirements.
- The layout of machines within each cell resembles a small assembly line.

Advantages of cellular layout

- Reduced material handling and transit time
- Reduced setup time
- Reduced work-in-process inventory
- Better use of human resources:

- Self-managed team, in most cases more satisfied with the work that they do
- Flexible resource. Workers in each cell are multifunctional and can be assigned to different routes within a cell or between cells as demand volume changes.
- Easier to control
- Easier to automate

Disadvantages of a cellular layout

- Inadequate part families

Part families identified for design purposes may not be appropriate for manufacturing purposes.

- Poorly balanced cells

It is more difficult to balance the flow of work through a cell than a single-product assembly line, because items may follow different sequences through the cell that require different machines or processing times.

- Expanded training and scheduling of workers

Training workers to do different tasks is expensive and time-consuming and requires the workers' consent.

- Increased capital investment

- Existing equipment may be too large to fit into cells or may be underutilized when placed in a single cell.

- Additional machines of the same type may have to be purchased for different cells.

- The cost and downtime required to move machines can also be high.

6.4.3 Production Flow Analysis (PFA)

Production flow analysis (PFA) is a well-established methodology used for transforming traditional functional layout into product-oriented layout. The method uses part routings to find natural clusters of workstations forming production cells able to complete parts and components swiftly with simplified material flow. Once implemented, the scheduling system is based on period batch control aiming to establish fixed planning, production and delivery cycles for the whole production unit. PFA is traditionally applied to job-shops with functional layouts, and after reorganization within groups lead times reduce, quality improves and motivation among personnel improves. PFA reduces non-value adding operations, introduces flow by evening out bottlenecks and diminishes process variability, all of which contribute to efficient operations management.

- **Production Flow Analysis (PFA)** is a group technology technique that reorders part routing matrices to identify families of parts with similar processing requirements
- The grouping into part families of items with similar design or manufacturing characteristics
 - Design characteristics: size, shape and function
 - Manufacturing or processing characteristics: type and sequence of operations required.

Production Flow Analysis (PFA) is a systematic methodology for comprehensive design of a factory using manual methods for machine grouping, part family formation, cell layout and shop layout design for industrial facilities. However, viewed in a larger context, it provides the framework that could be used by a company to identify reasons for chaotic material flows in its facility. The power of PFA is due to its applicability at different levels of resolution in any factory, i.e. flows between buildings, flows between departments or shops within the same building, flows between machines or work-centers or cells in a shop, flows between machines in a manufacturing cell, flows of tools in the tool magazine of a machining center, movements of an operator at a workstation, etc.

When applied to a single factory, the classic framework for manual implementation of PFA consists of four stages, each stage achieving material flow reduction for a progressively reducing portion of the factory: Factory Flow Analysis (FFA), Group Analysis (GA), Line Analysis (LA) and Tooling Analysis (TA).

7.0 JUST-IN-TIME (JIT) PRINCIPLES

'Just-in-time' is a management philosophy and not a technique.

It originally referred to the production of goods to meet customer demand exactly, in time, quality and quantity, whether the 'customer' is the final purchaser of the product or another process further along the production line.

When items are ready just in time, they don't occupy so much space.

Rhetoric of Just In Time

When is the best time to have an inventory part ready for production? Just in time.

When is the best time to have an item ready for the next step in production? Just in time.

When is the best time to have a product ready for delivery to a customer? Just in time.

So why do manufacturers build inventory of both finished goods and raw materials? *Just in case!*

A buffer of inventory on hand is comforting – and costly. If you hold a lot of items in inventory, you're locking away a huge amount of cash unnecessarily. These items can be lost, stolen, or damaged, or they can deteriorate. They occupy space, which could otherwise be devoted to operations. And they can become obsolete, particularly when products are improved or changed often (many of us can remember images of airfields full of 'unwanted, obsolete cars from the 1970s and 1980s.) All of this represents financial loss to the business.

In the 1970s, when Japanese manufacturing companies were trying to perfect their systems, Taiichi Ohno of Toyota developed a guiding philosophy for manufacturing that minimized waste and improved quality. Called Just In Time (JIT), this philosophy advocates a lean approach to production, and uses many tools to achieve this overall goal.

When items are ready just in time, they aren't sitting idle and taking up space. This means that they aren't costing you anything to hold onto them, and they're not becoming obsolete or deteriorating. However, without the buffer